

Statistics for Insurance — Chapter 1

Solutions to Exercises 9–25 and the Practical Exercises

Exercise 9. Functional CLT for the Conditional Cumulative Hazard

Define

$$\Lambda^{(n)}(t \mid x) = -\log(1 - \widehat{F}_n(t \mid x)), \quad \Lambda(t \mid x) = -\log(1 - F(t \mid x)),$$

and write

$$c_K^2 = \frac{R_K}{f(x)}.$$

Theorem

Assume the conditions of Theorem 2.8, and let $\tau > 0$ satisfy

$$F(\tau \mid x) < 1.$$

Then, in $\ell^\infty([0, \tau])$,

$$\sqrt{nh} (\Lambda^{(n)}(\cdot \mid x) - \Lambda(\cdot \mid x)) \xrightarrow{d} c_K \frac{B(F(\cdot \mid x))}{1 - F(\cdot \mid x)},$$

where B is a standard Brownian bridge. The limit is a centred Gaussian process with covariance

$$\text{Cov}(s, t) = \frac{R_K}{f(x)} \frac{F(s \wedge t \mid x)}{1 - F(s \wedge t \mid x)}.$$

Proof

Step 1 (Weak convergence of the cdf). Restriction from $[0, \infty)$ to $[0, \tau]$ is continuous. Hence, by Theorem 2.8 and the continuous mapping theorem,

$$\sqrt{nh} (\widehat{F}_n(\cdot \mid x) - F(\cdot \mid x)) \xrightarrow{d} c_K B \circ F(\cdot \mid x) \quad \text{in } \ell^\infty([0, \tau]).$$

Step 2 (Hadamard differentiability). Let

$$p = F(\tau \mid x) < 1$$

and define

$$\varphi(y) = -\log(1 - y)$$

on

$$\left(-1, \frac{1+p}{2}\right).$$

On this interval,

$$\varphi'(y) = \frac{1}{1-y} \leq \frac{2}{1-p}, \quad \varphi''(y) = \frac{1}{(1-y)^2} \leq \frac{4}{(1-p)^2}.$$

Thus φ' is bounded and Lipschitz, hence uniformly continuous. Since $F(\cdot \mid x)$ takes values in $[0, p]$, Lemma 1.10(2) shows that

$$\phi(G) = \varphi \circ G$$

is Hadamard differentiable at $F(\cdot \mid x)$, with derivative

$$\phi'_F(g) = \frac{g}{1 - F(\cdot \mid x)}.$$

Step 3 (Functional δ -method). By Step 1,

$$\sup_{t \in [0, \tau]} \left| \widehat{F}_n(t \mid x) - F(t \mid x) \right| \xrightarrow{p} 0.$$

Hence $\widehat{F}_n(\cdot \mid x)$ lies in the domain of φ with probability tending to one. Theorem 1.17 gives

$$\sqrt{nh} (\Lambda^{(n)} - \Lambda) \xrightarrow{d} \phi'_F(c_K B \circ F) = c_K \frac{B(F(\cdot \mid x))}{1 - F(\cdot \mid x)}.$$

Step 4 (Covariance). Using

$$\text{Cov}(B(u), B(v)) = u \wedge v - uv,$$

let

$$F_s = F(s \mid x), \quad F_t = F(t \mid x).$$

For $s \leq t$,

$$\begin{aligned} \text{Cov}(s, t) &= c_K^2 \frac{F_s - F_s F_t}{(1 - F_s)(1 - F_t)} \\ &= c_K^2 \frac{F_s}{1 - F_s} \\ &= c_K^2 \frac{F(s \wedge t \mid x)}{1 - F(s \wedge t \mid x)}. \end{aligned}$$

Since $c_K^2 = R_K/f(x)$,

$$\boxed{\text{Cov}(s, t) = \frac{R_K}{f(x)} \frac{F(s \wedge t \mid x)}{1 - F(s \wedge t \mid x)}}. \quad \blacksquare$$

Remarks

- The condition $F(\tau \mid x) < 1$ is essential. It keeps φ' bounded, so Lemma 1.10(2) applies, and it keeps the variance finite.
- For fixed t , the result reduces to the Chapter 1 pointwise result for the cumulative hazard, with two changes: $\sqrt{n}$ becomes $\sqrt{nh}$, and the variance gains the factor $R_K/f(x)$. That is,

$$\sqrt{nh} (\Lambda^{(n)}(t \mid x) - \Lambda(t \mid x)) \xrightarrow{d} N \left(0, \frac{R_K}{f(x)} \frac{F(t \mid x)}{1 - F(t \mid x)} \right).$$